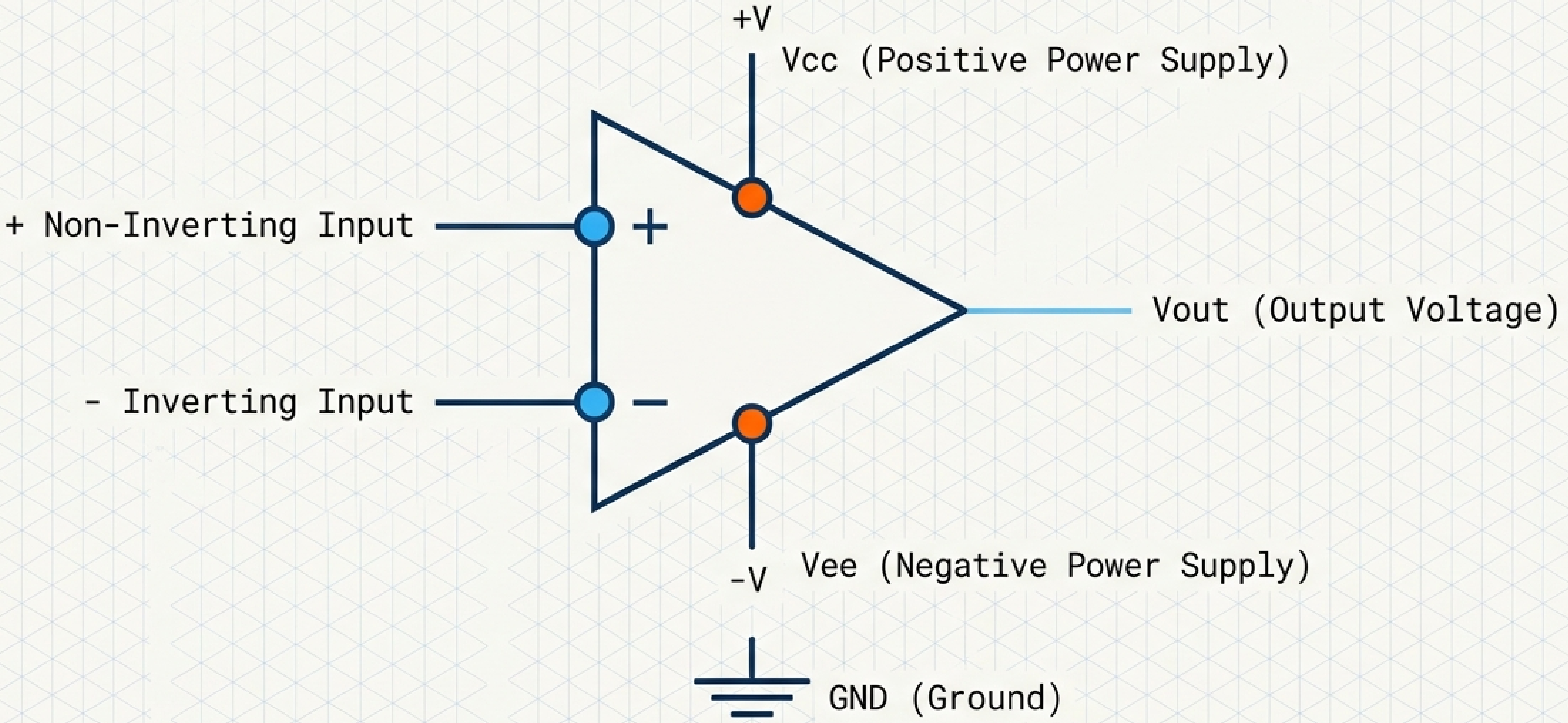
The Architecture of Amplification

A Structural Framework for Electronic Energy Control



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The Illusion of the Voltage Multiplier

An amplifier is not a passive multiplier. It is a signal transformation system relying on device parameters, load conditions, and external power.

The Myth

$y(t) = A\,x(t)$

(Only valid in limited, ideal linear states)

The Reality

Input Signal

Power Supply Energy

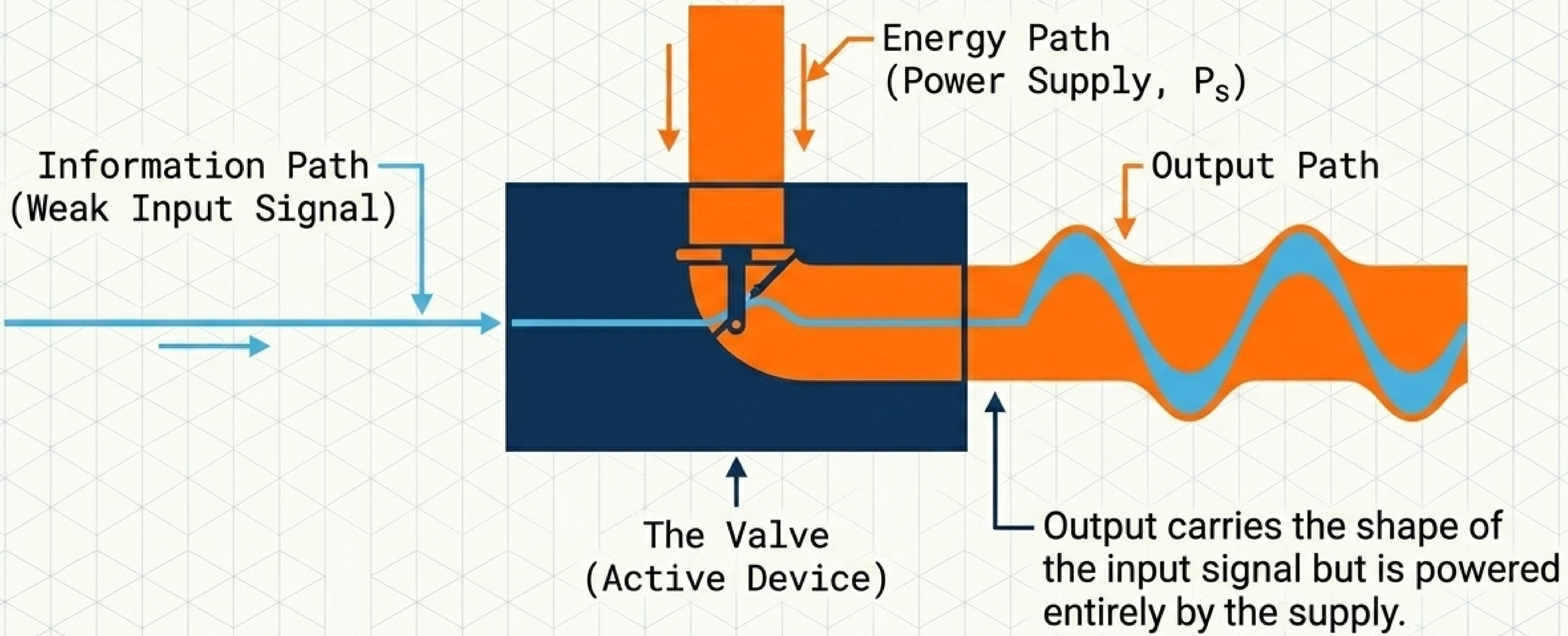
$$\boxed{y(t) = F(x(t), P_S, \theta, L, t)}$$

Load Condition

Device Parameters

The Energy Valve Model

Amplification is not the creation of energy; it is the controlled conversion of supply energy, regulated by an input signal.



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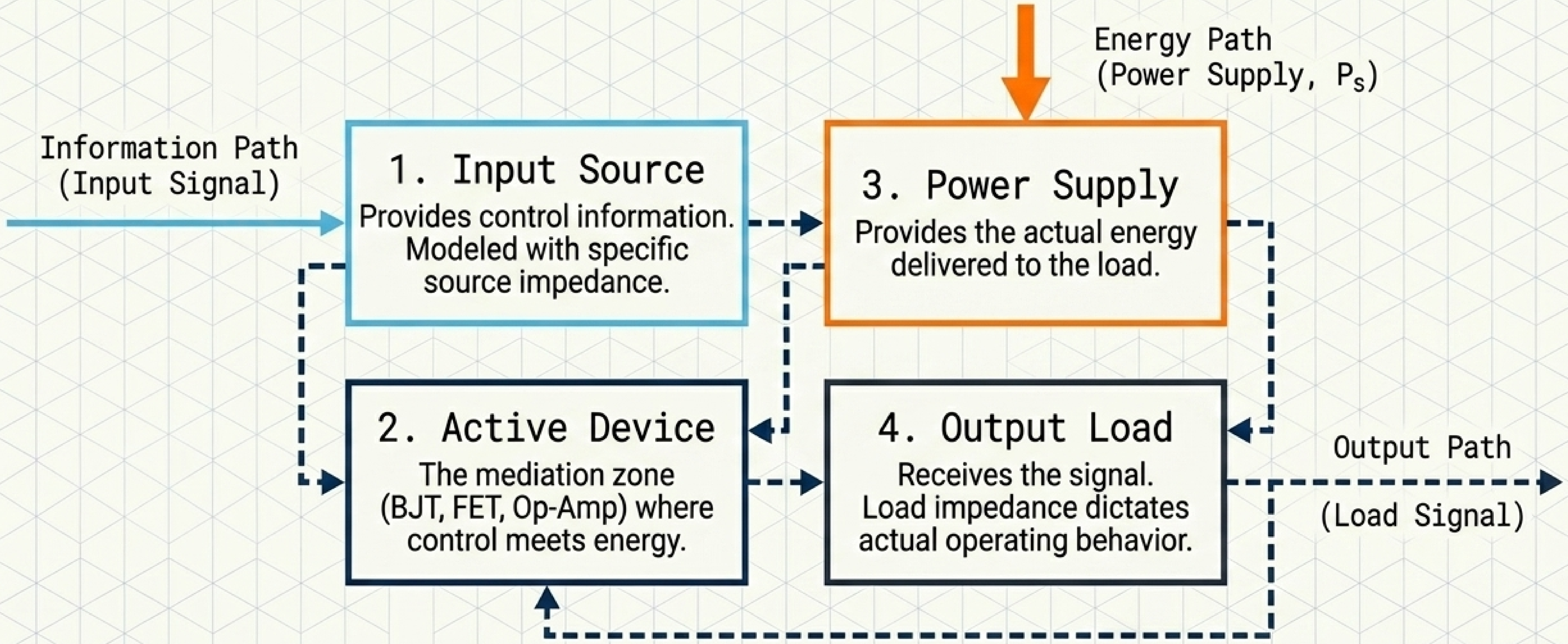
The Four Inseparable Elements of Amplification

No amplifier can be completely understood by examining only its input and output voltages. A change in the load or supply fundamentally alters the active device's behavior.



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The Transfer Function Matrix

Amplification is not limited to voltage. A buffer or transresistance amplifier is equally mathematically valid.

Voltage Gain

$$A_v = \frac{V_o}{V_i}$$

$$G_v = 20\log_{10}|A_v|$$

Current Gain

$$A_i = \frac{I_o}{I_i}$$

$$G_i = 20\log_{10}|A_i|$$

Power Gain

$$A_p = \frac{P_o}{P_i}$$

$$G_p = 10\log_{10}|A_p|$$

Transconductance

$$g_m = \frac{I_o}{V_i}$$

Transresistance

$$r_m = \frac{V_o}{I_i}$$

Buffer Amplifier

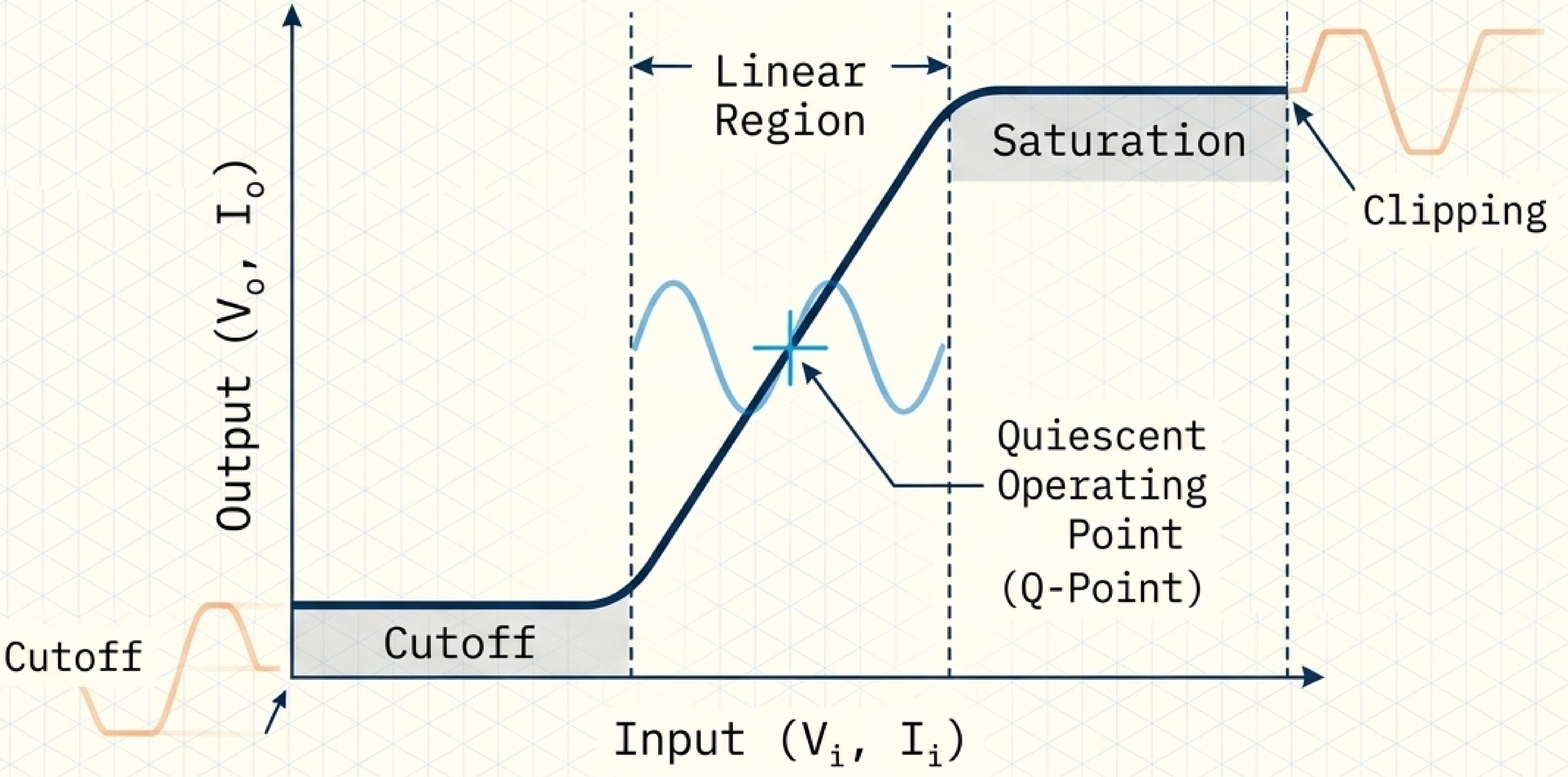
Unity gain ($A \approx 1$)

$$Z_{out} = \frac{v_o}{i_o}$$

Provides lower output impedance and current drive.

The Biasing Prerequisite: Targeting the Q-Point

Amplification requires establishing a quiescent state. The small-signal model only exists because the bias point allows local linear approximations of nonlinear device physics.



The Multidimensional Classification Matrix

Operating class does not define an amplifier. A Class AB amplifier may be an audio power stage or an RF transmitter. Classification must encompass all six axes.

1. FUNCTION	2. TOPOLOGY	3. OPERATING CLASS	4. FREQUENCY	5. FEEDBACK	6. APPLICATION
Voltage	Common Emitter	Class A, B, AB	DC	Open-loop	Low Noise
Current	Differential	Class D, E, F	Audio	Global Negative	Instrumentation
Power	Cascode	(Defines conduction/ switching behavior only)	RF	Feedforward	Biomedical
Transconductance	Multistage		Microwave		Power Control
Buffer					

The Analytical Boundaries: Small vs. Large Signal

Small signal gain describes local behavior. Large signal analysis describes the real amplitude limits of physical engineering.

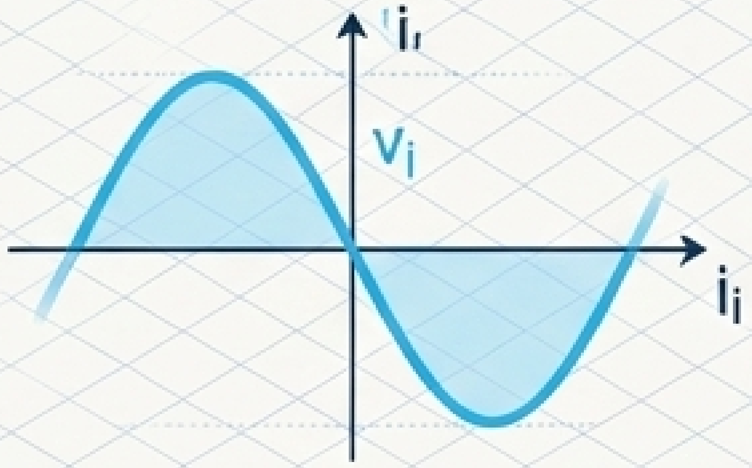
Small Signal Domain
(Local Linear Behavior)

Governing Math: $A(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$

Key Metrics:
- Voltage Gain (A_v)
- Input Impedance ($Z_{in} = v_i / i_i$)
- Small-signal Bandwidth

Assumptions:

Device operates purely within the linear region near the bias point.



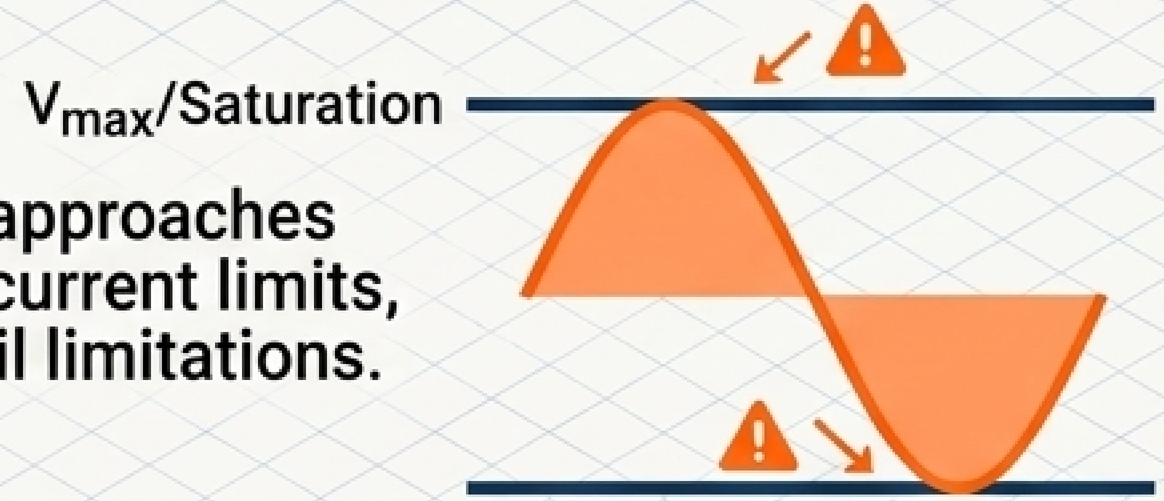
Large Signal Domain
(Physical Constraints)

Governing Math: $SR \geq 2\pi f\, V_p$

Key Metrics: Slew Rate → Clipping
- → Compression
- → Thermal Limits

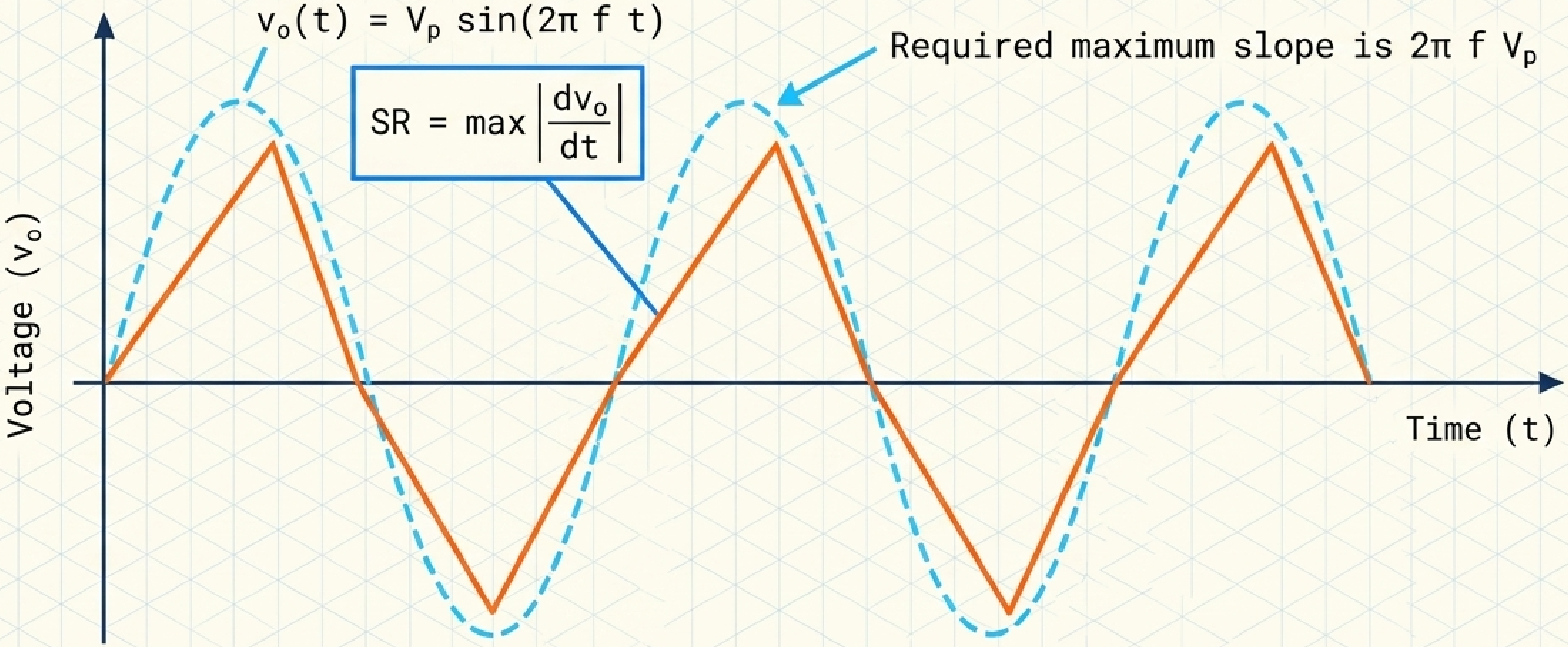
Reality:

The device approaches saturation, current limits, or supply rail limitations.



The Slew Rate vs. Bandwidth Paradox

If $2\pi f V_p > SR$, the amplifier turns a sine wave into a slew-limited triangle wave. Bandwidth without sufficient Slew Rate guarantees distortion under large signal conditions.



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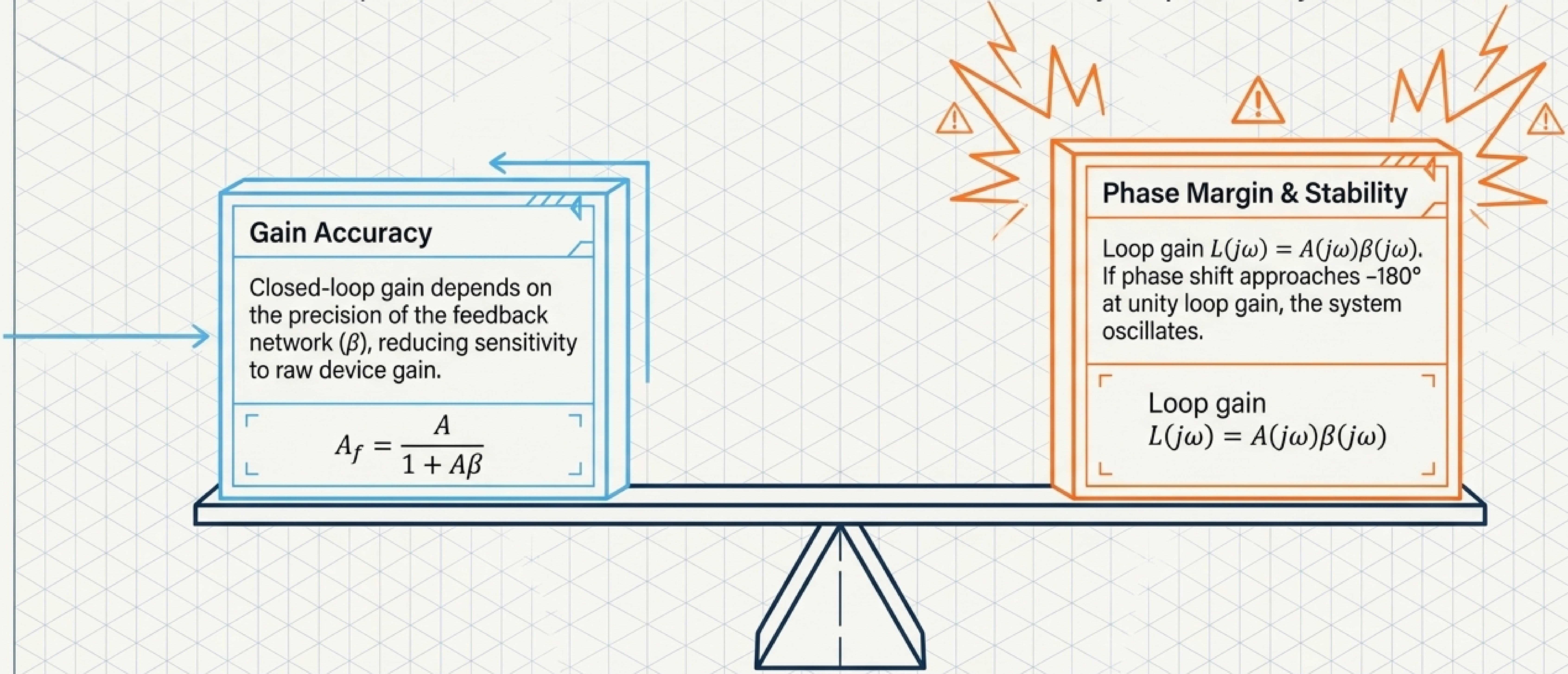
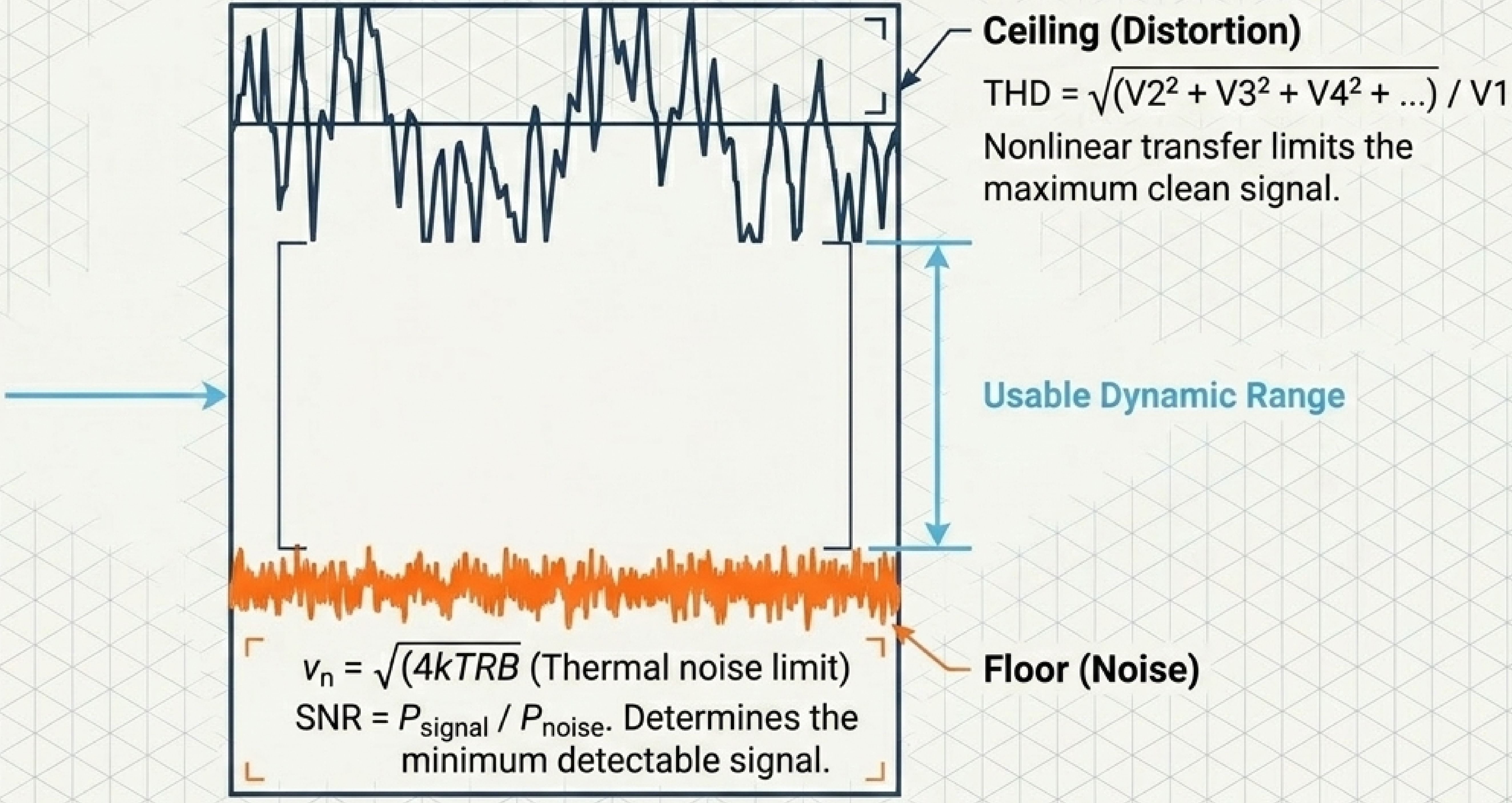
The Feedback Loop Scale: Accuracy vs. Stability

Feedback is both a powerful correction mechanism and a strict stability responsibility.



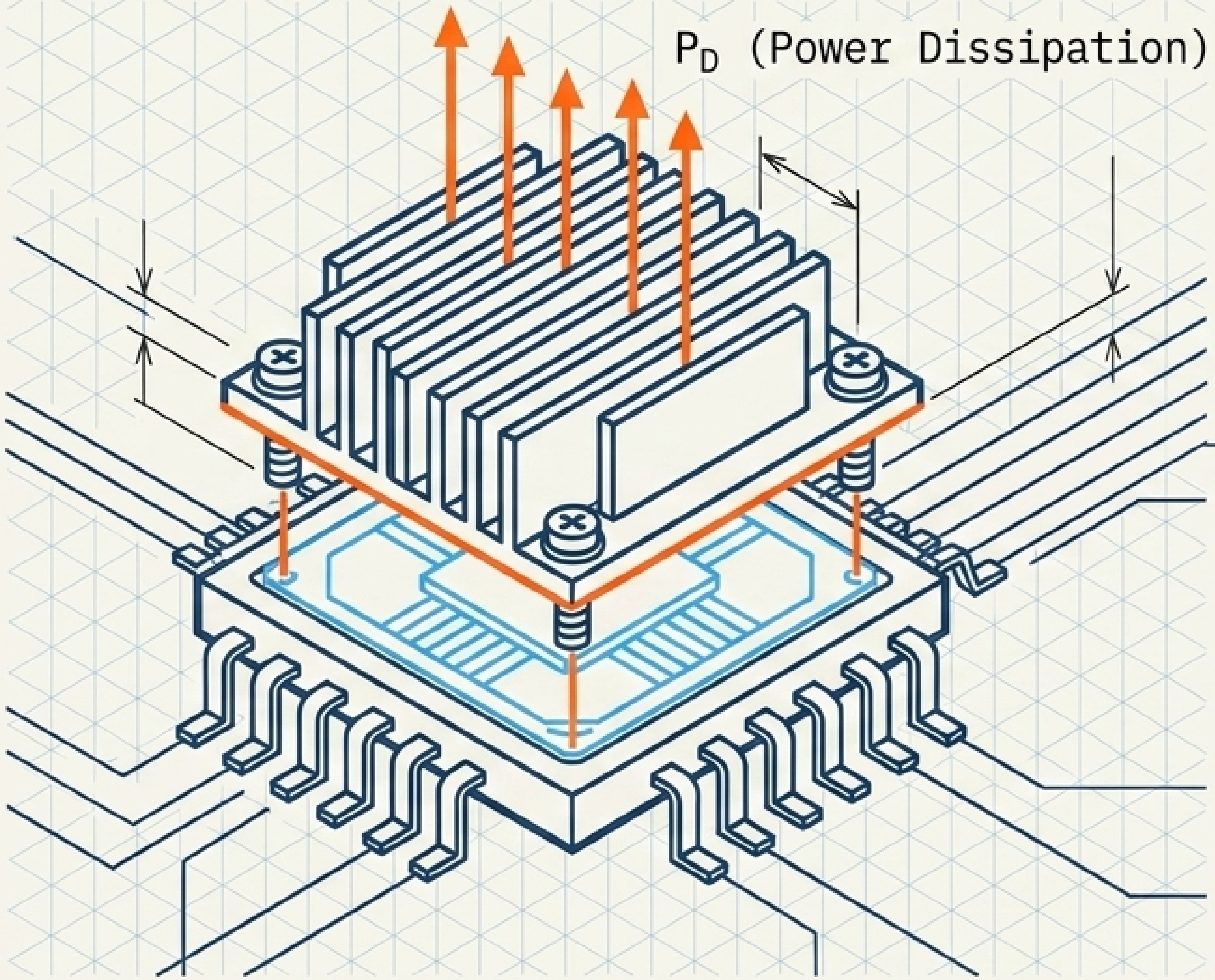
Signal Impurity: The Floor and The Ceiling

The output is never a pristine copy of the input. The usable signal window is squeezed between thermal noise and nonlinear distortion.



The Thermal Boundary: Efficiency as a Constraint

Heat is not a secondary inconvenience. It changes device parameters, limits output power, and causes thermal runaway. Electrical and thermal design are inseparable.



Efficiency

$\eta = \dfrac{P_o}{P_{dc}}$ (How much supply power becomes useful output).

Dissipation

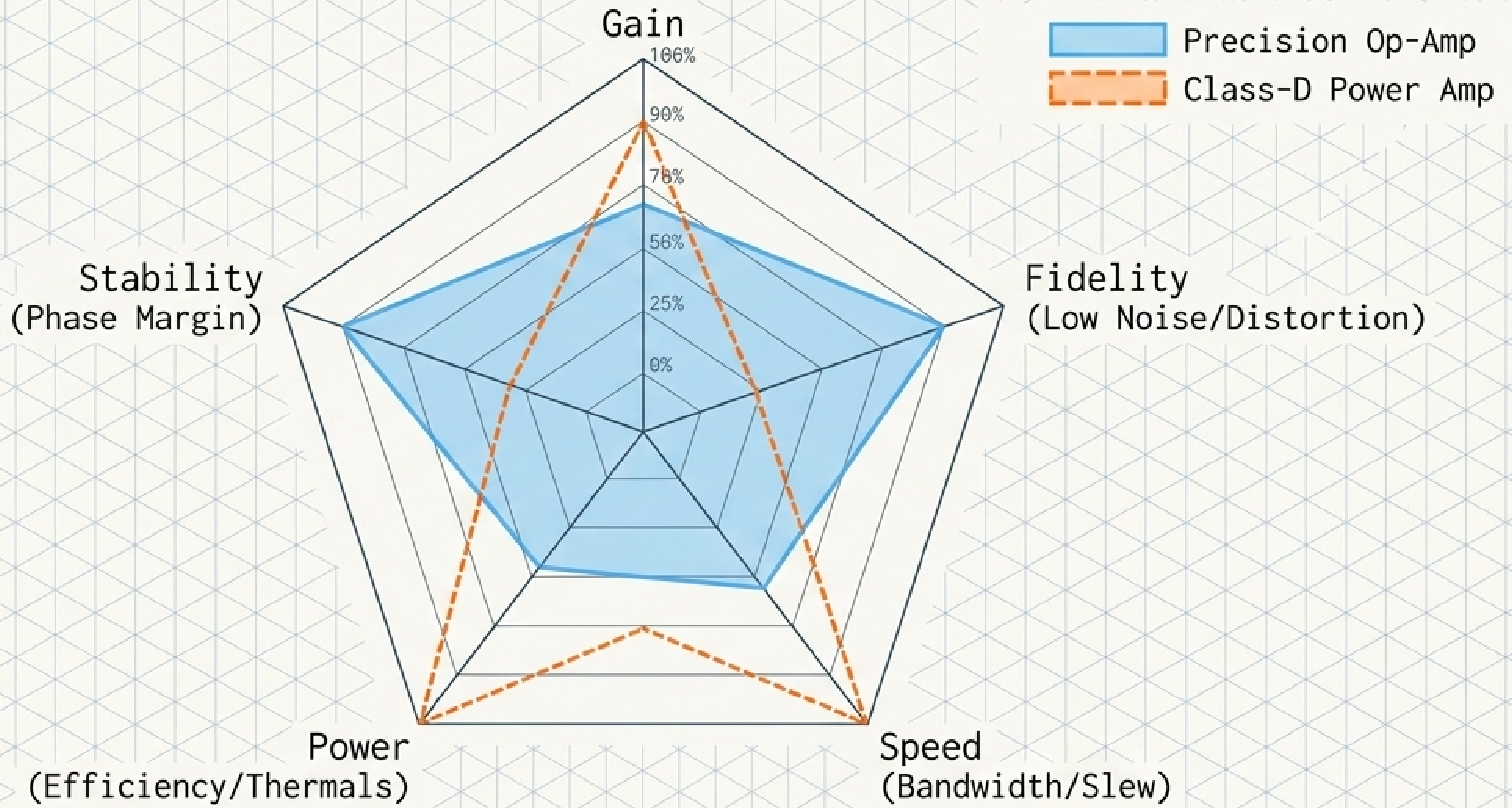
$P_D = P_{dc} - P_o$ (The remainder converts to heat).

Thermal Rise

$\Delta T = P_D R_\theta$ (Where R_θ is thermal resistance).

The Illusion of the Single Metric

A high-gain amplifier is not inherently a good amplifier. Real engineering requires disciplined optimization across interdependent vectors.



The Architectural Evaluation Framework

Judgment is only valid after identity, classification, and analysis have been completed. This is the structure of electronic evaluation.

4. **Evaluation & Validation:**
Test the **physical constraints**.
(Evaluate noise floor, thermal rise, load interaction, and stability margins.)

3. **Analysis:** Map **mathematical behavior**.
(Calculate Av, bandwidth, impedance, slew rate, distortion.)

2. **Classification:** Map the **6 axes**.
(Function, Topology, Class, Frequency, Feedback, Application.)

1. **Identity:** Establish the object. (Is it active? Does input control supply energy?)